

Thesis Defence
4th of December 2025

Doctoral School MaSTIC (N°641)

Automatic Text Revision for Scientific Writing Assistance

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Supervision : Richard Dufour, Florian Boudin & Nicolas Hernandez



Context

Domain

- Scientific writing assistance
- Focus on the revision step
- Applied to scientific writing

Motivations

- Writing an article is challenging
- Strong writing skills are essential
- Especially difficult for junior researchers and non-native English speakers

What is Scientific writing?

Academic writing

- **Context** : University setting
- **Form**: Essay, thesis, syllabus, articles ...



Scientific writing

- **Form**: articles published in journals or conferences
- **Specificities**:
 - mostly written in English
 - concise, precise, clear
 - codified usage of tenses, pronouns and terminology
- **Structure**: IMRaD format
 - Introduction, Related work, Methods, Results, Discussion, Conclusion



Writing assistance Tasks



bibliographic
management



argument mining



grammatical
error correction



stylometric analysis



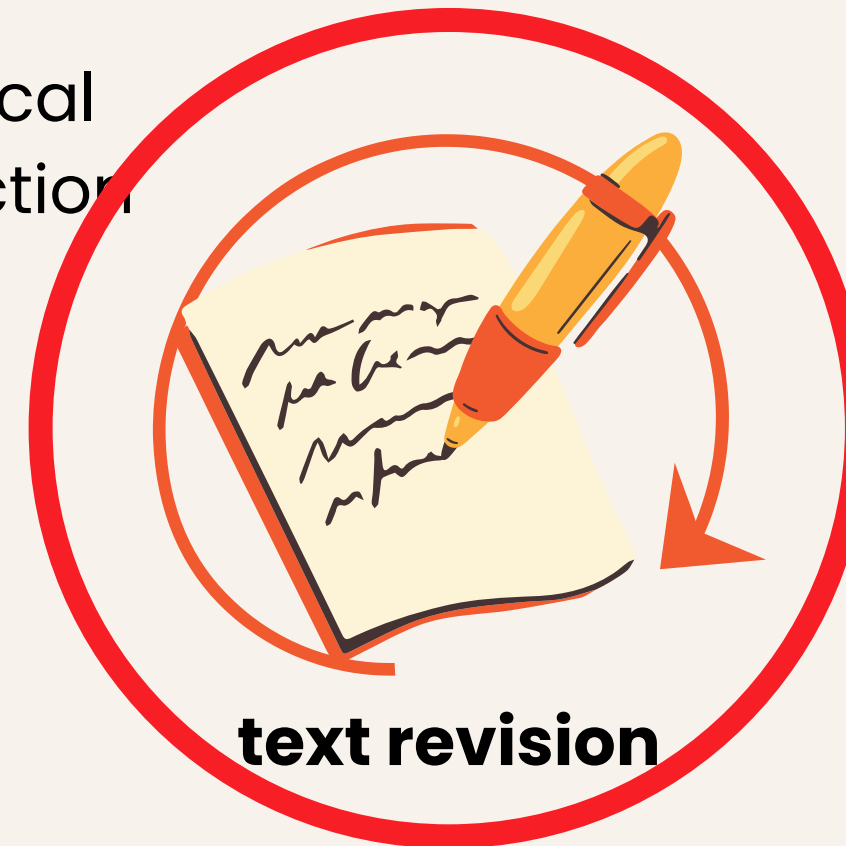
thematic edit
visualisation



semantic relation
extraction



prediction of paper
acceptance/rejection



citation
recommendation

Research questions

How to **define** the task to observe substantial revisions?

How to construct a **dataset** that allow to capture revisions?

Which metrics align best with human judgements to **evaluate** STR?

Task definition

Resources

Evaluation methods

Scientific Text Revision = **STR**

Building a framework for STR

I – Task definition

Revision definition

Writing process

Sentence vs paragraph level

II – Building Resources

CASIMIR

Creation

Data analysis

ParaRev

Creation

Statistics

III – Identifying Reliable Evaluation Metrics

Limitations of traditional metrics

Exploring alternative metrics

Metrics recommendation

Text revision in Scientific Writing Assistance: An Overview

1st author, BIR 2023 (Workshop at ECIR 2023)

ParaRev : Building a dataset for Scientific Paragraph Revision annotated with revision instruction

1st author, collaboration with NII, WRAICOGS 2025 (Workshop at COLING 2025)



The 31st International
Conference on Computational
Linguistics

Revision task

Definition

Text revision is the transformation of an input text into an improved version fitting a desired attribute (formality, clarity, etc.), closer to the intended text

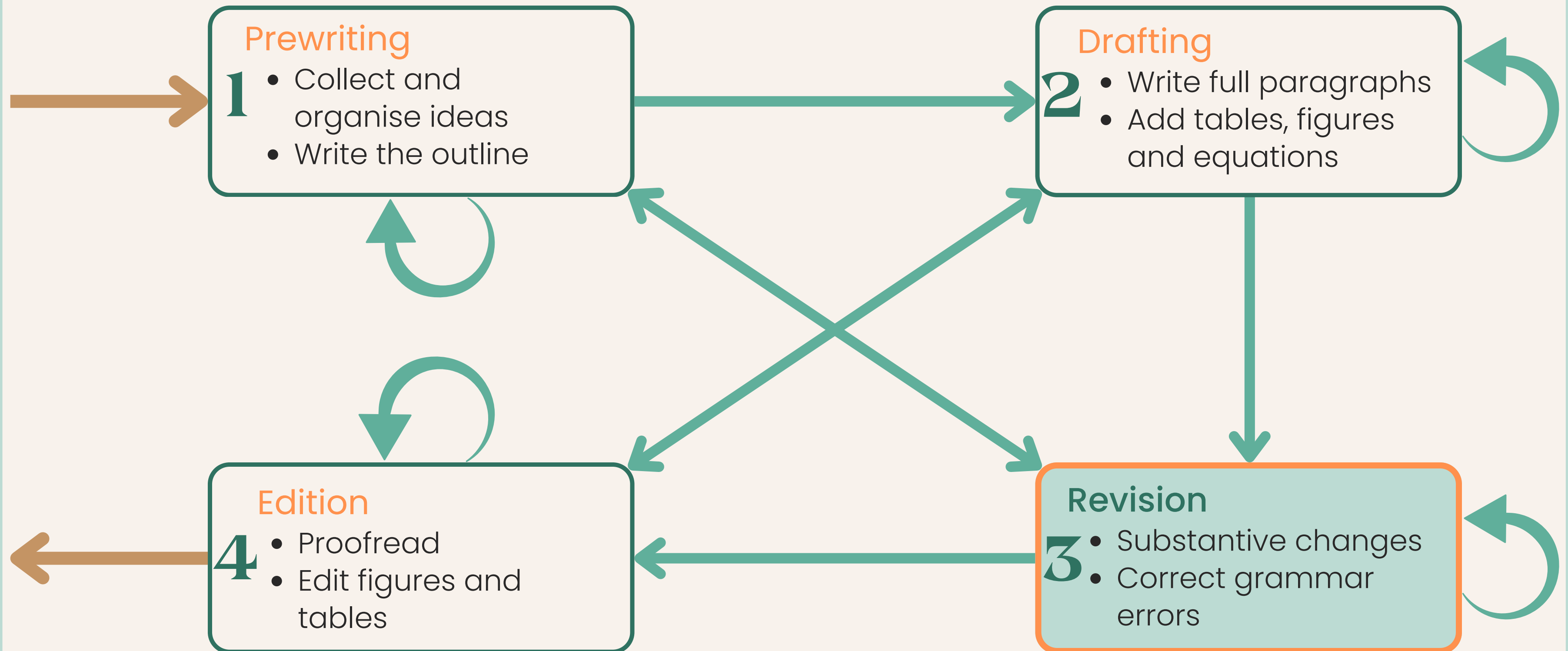
Example

The model has good results.

Our model shows good results in this task.

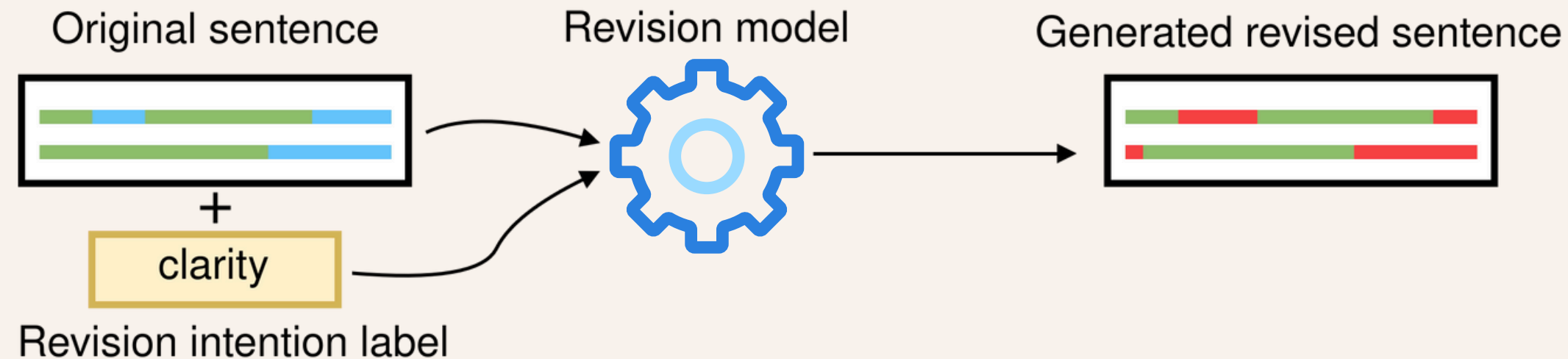
Our model shows excellent performance in this task.

The writing process of scientific articles

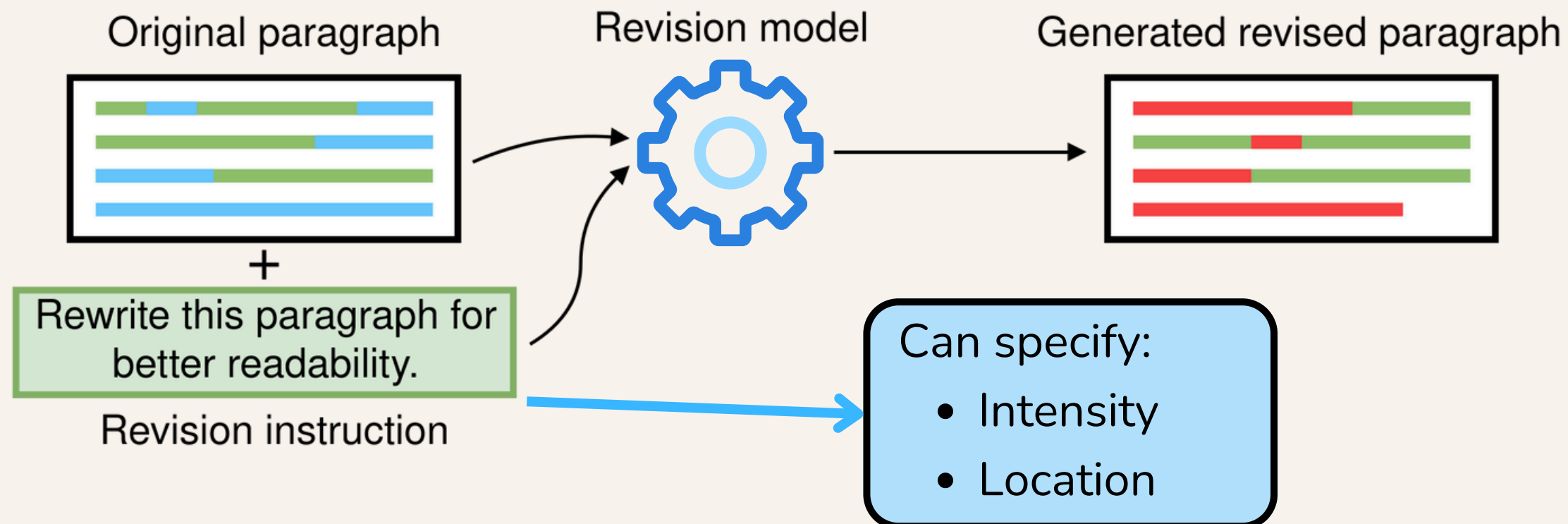


Sentence vs paragraph revision tasks

Sentence revision: Traditional definition



Paragraph revision: Proposed definition



Example of a revised paragraph

Source paragraph

More recently, the focus **has** shifted to **representation** learning in a latent space, e.g. through contrastive losses (van den Oord et al., 2018; He et al., 2020; Chen et al., 2020), negative-free similarity (Grill et al., 2020; Chen & He, 2021), online clustering and self-distillation (Caron et al., 2020; 2021; Li et al., 2021). **For all these methods, data augmentation is a key component.** In particular, **we regard the popular** multi-crop **strategy** as **a crude** suggestion of objectness, as it **implicitly** encourages learning similar representations between the global view of an object and **a cropped view of** its parts. **Although recent extensions introduce patch-based contrastive learning (Li et al., 2021; Wang et al., 2021), the resulting features remain redundant. How to summarize dense features into sparse representations is an area of active research (Hinton, 2021).**

Revised paragraph

Then, the focus shifted to learning in a latent space, e.g. through contrastive losses (van den Oord et al., 2018; He et al., 2020; Chen et al., 2020), negative-free similarity (Grill et al., 2020; Chen & He, 2021), online clustering and self-distillation (Caron et al., 2020; 2021; Li et al., 2021). **Recent works formulate patch-based losses (Li et al., 2021; Wang et al., 2021), yet the resulting embeddings remain redundant. How to summarize dense features into a sparse representation is an active area of research (Hinton, 2021). A key components of all these methods is data augmentation.** In particular, multi-crop **can be regarded** as **an implicit** suggestion of objectness, as it encourages learning similar representations between the global view of an object and its **cropped** parts.

Paragraph Revision taxonomy

Rewriting	Light	Changes in the choice of words
	Medium	Complete rephrasing of sentences
	Heavy	Complete rephrasing of the paragraph
Concision		Same idea, stated more briefly. Details are deleted
Development		Same idea, stated out at greater length by adding details or definitions of the terms used
Content	Addition	Modification on the content — Addition of a new idea
	Substitution	Modification on the content — Substitution of an idea or a fact by an other
	Deletion	Modification on the content — Deletion of an idea
Unusable	Segmentation problems (Footnote mixed with text), misalignment (paragraphs that have nothing to do with each other) and others problems coming from document processing	

Chapter recap

- Definition of the **writing process in 4 steps**
 - *Prewriting*
 - *Drafting*
 - **Revision**
 - *Edition*
- Formal definition of the text revision task at **paragraph-level**, with personalised **revision instructions**
 - An associated revision taxonomy of 9 categories

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CASIMIR: A Corpus of Scientific Articles Enhanced with Multiple Author-Integrated Revisions

1st author, LREC-COLING 2024 + workshop ARTS

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LREC-COLING 2024

**CORIA
TALN** 2023
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Linguistics

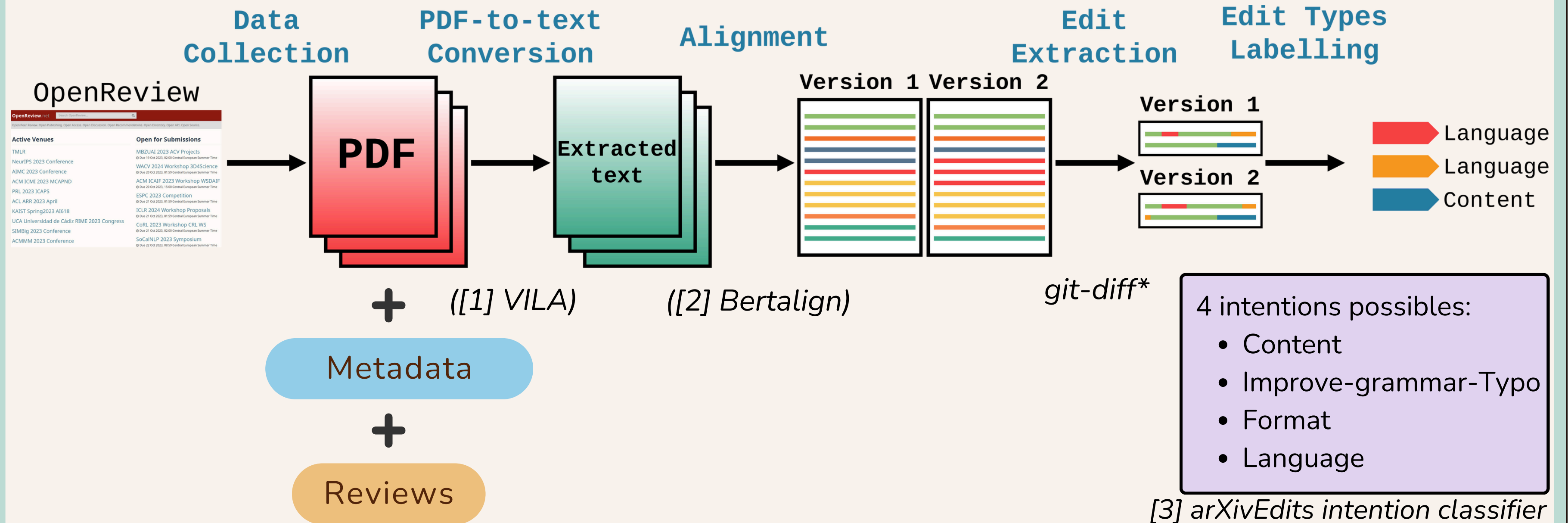
Datasets

Beginning of creation → Published

		Multiple versions (>2)	Authentic revisions	Full length	Granularity (Section/Parag)	Early versions	Size (<1k/<10k/>10k)	Annotated
CASIMIR ParaRev	Cornell Statement Strength	✗	✓	✗	✗	✗	✓	✗
	SMITH	✗	✗	✗	✗	✗	✓	✗
	IteraTeR	✓	✓	✗	✓	✗	✓	✓
	arXivEdits	✓	✓	✓	✓	✗	✗	✓
	F1000RD	✓	✓	✓	✓	✓	👉	✗
	ARIES	✗	✓	✓	✓	✓	👉	👉
	NLPeer	✓	✓	✓	✓	✓	👉	✗
	MOPRD	✓	✓	✓	✓	✓	👉	✗
	TETRA	✗	✗	✗	✓	✗	✗	✓
	KIWI	✓	✗	✗	✓	✓	✗	✓
	ScholaWrite	✓	✓	✓	✓	✓	✗	✓

Characteristics of previous datasets for scientific text revision

Creation of the casimir corpus



Content

Article pairs

- 15,646 different articles
- 3.5 versions by articles on average
- 36,733 pairs of versions



CASIMIR

Reviews

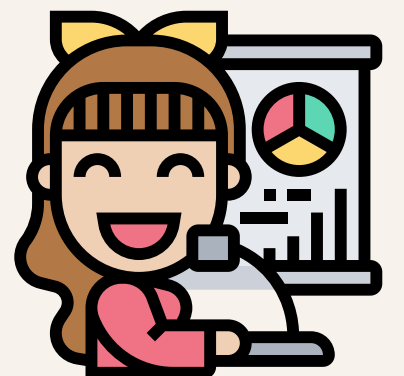
- Comments (can contain grades)
- Acceptance decision
- Dates...

Metadata

- Dates
- Authors
- Keywords
- Conference
- Ids...

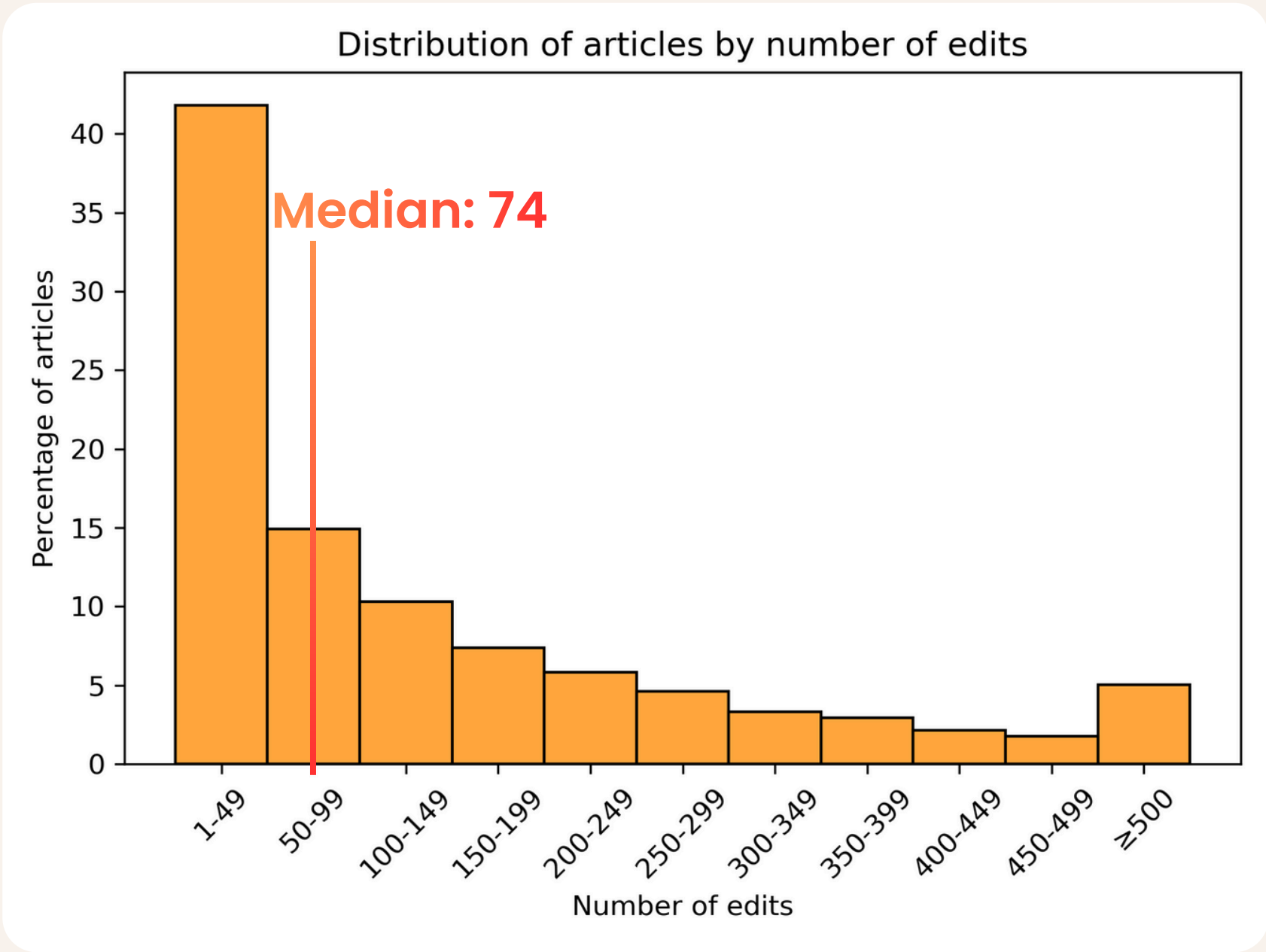
29 conferences

Domains : machine learning (ICLR, ICML, NeurIPS), robotics (RSS, CoRL), NLP (ACL) and computer vision (ECCV)



Distribution of edits

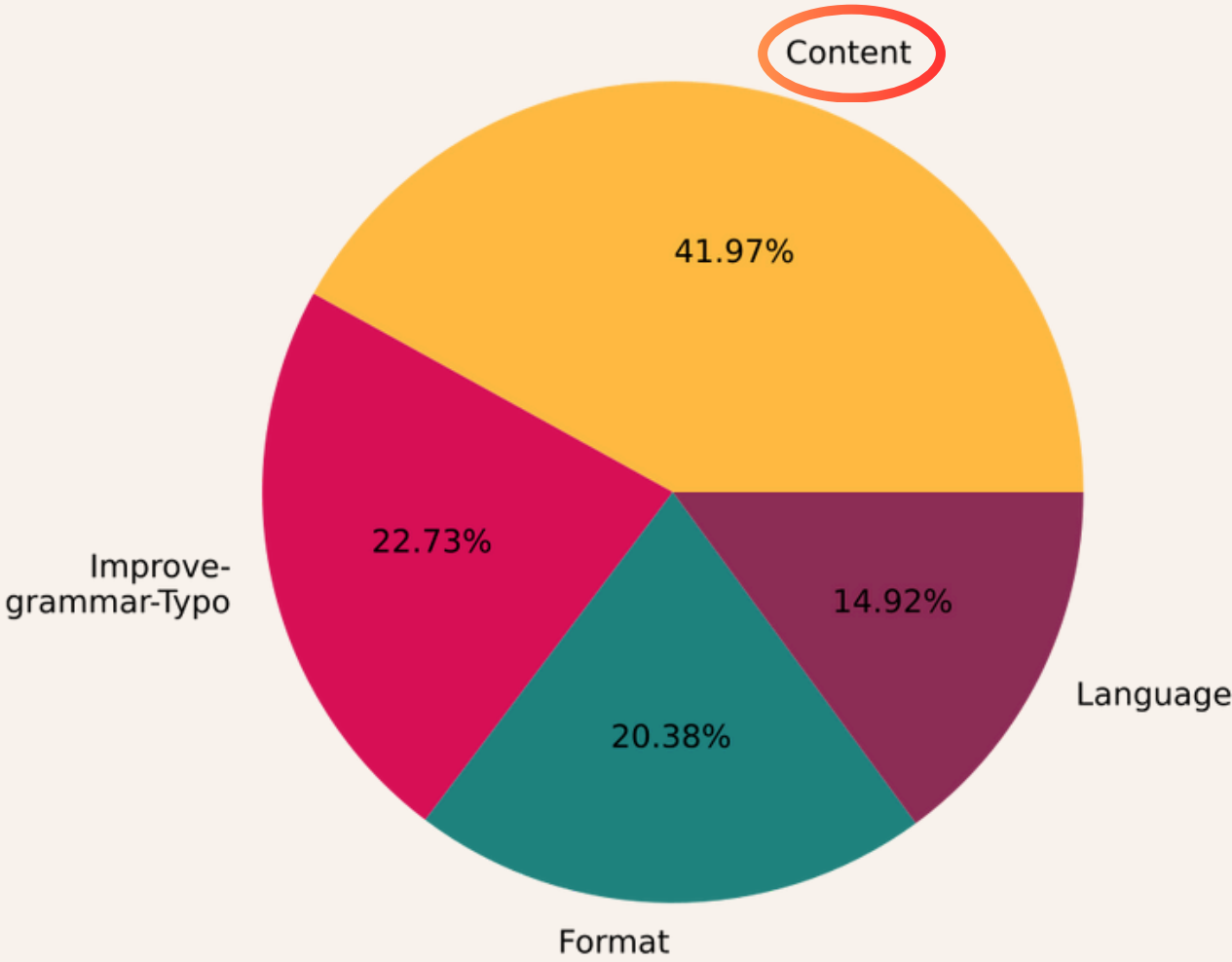
5.2M of individual **edits** distributed in 3.7M of edited sentences



Distribution of articles by number of **edits**

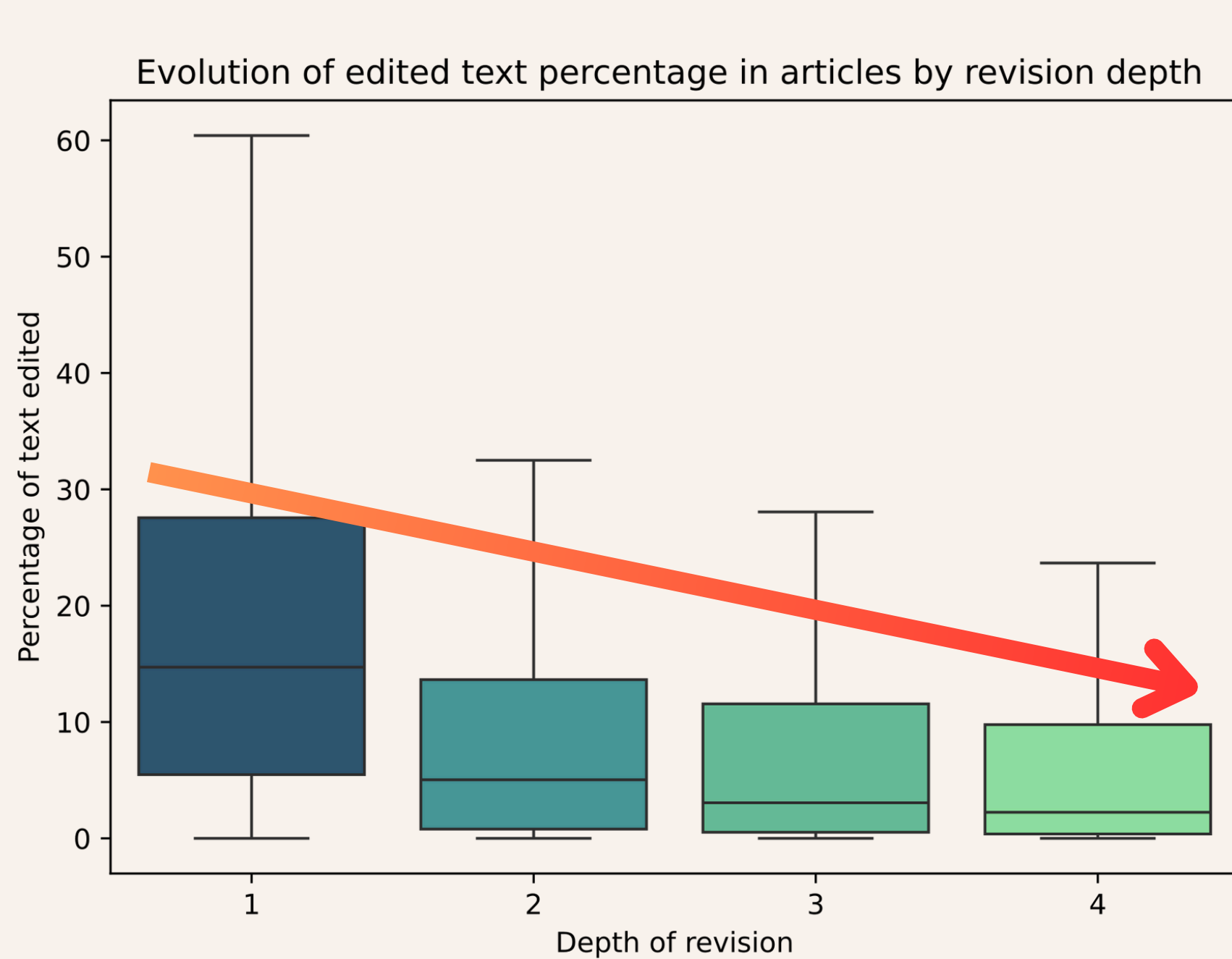
	Edits length
Median	13
Avg	34.88

Distribution of the **edits length**

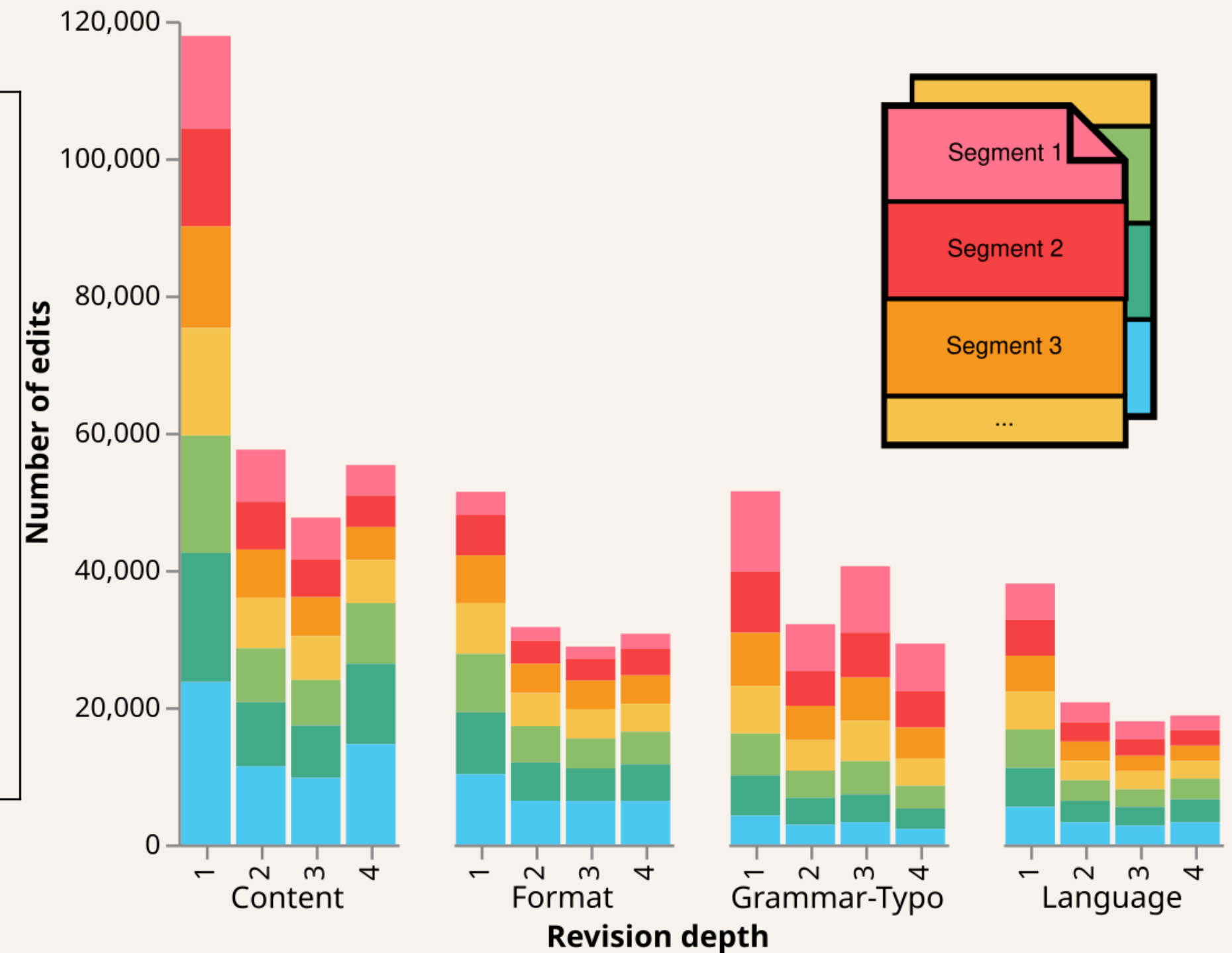


Distribution of **edit intentions**

Evolution and location of edits



Evolution of edited text percentage
in articles by revision depth



Evolution of the location of edited text
by intention and revision depth

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**CORIA
TALN** 2023
RUCRI | REGITAL

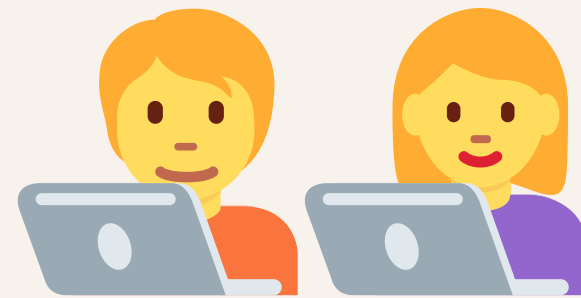

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ParaRev Data pipeline – Annotation

Manual Annotation (641)

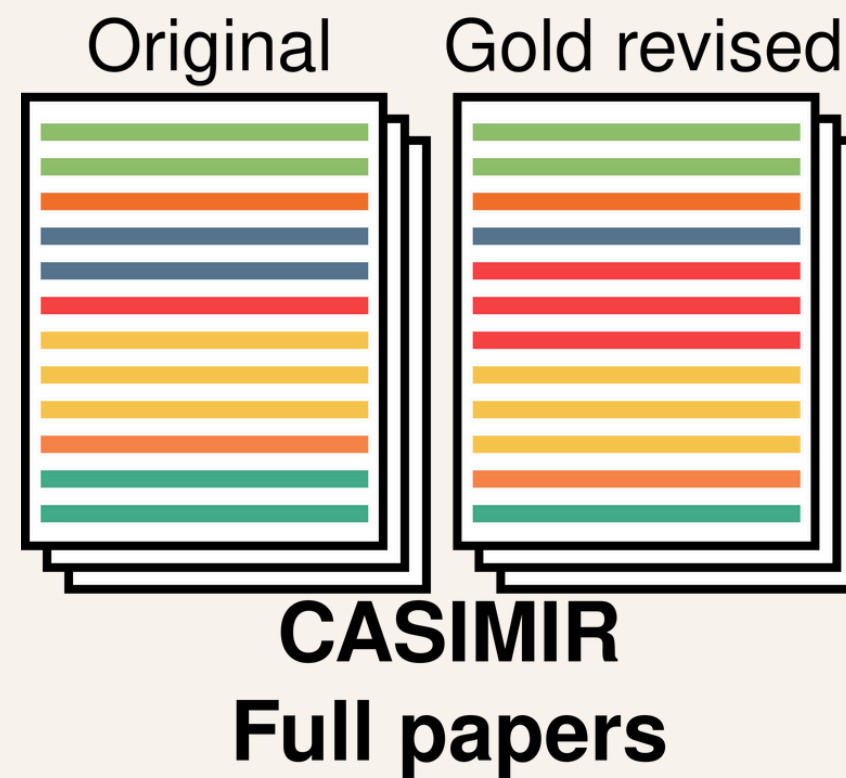
Manual annotation



Label

Instruction

- Overall intention
- Up to 2 labels



1.8M modified paragraphs



Interesting paragraphs extraction (48,203)

Instruction

When is an instruction provided?



A paragraph have an associated instruction only when “Development”, “Content addition” and “Content substitution” are **not part** of the list of intentions.

How is an instruction written?



Instructions are **simple and concise**.



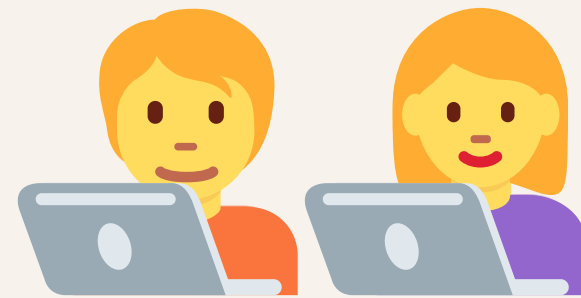
Instructions can be used to lead the model on the **location of the modifications**.



In real world usage, a paragraph can be **revised on a specific portion** and the rest serve as **context**.

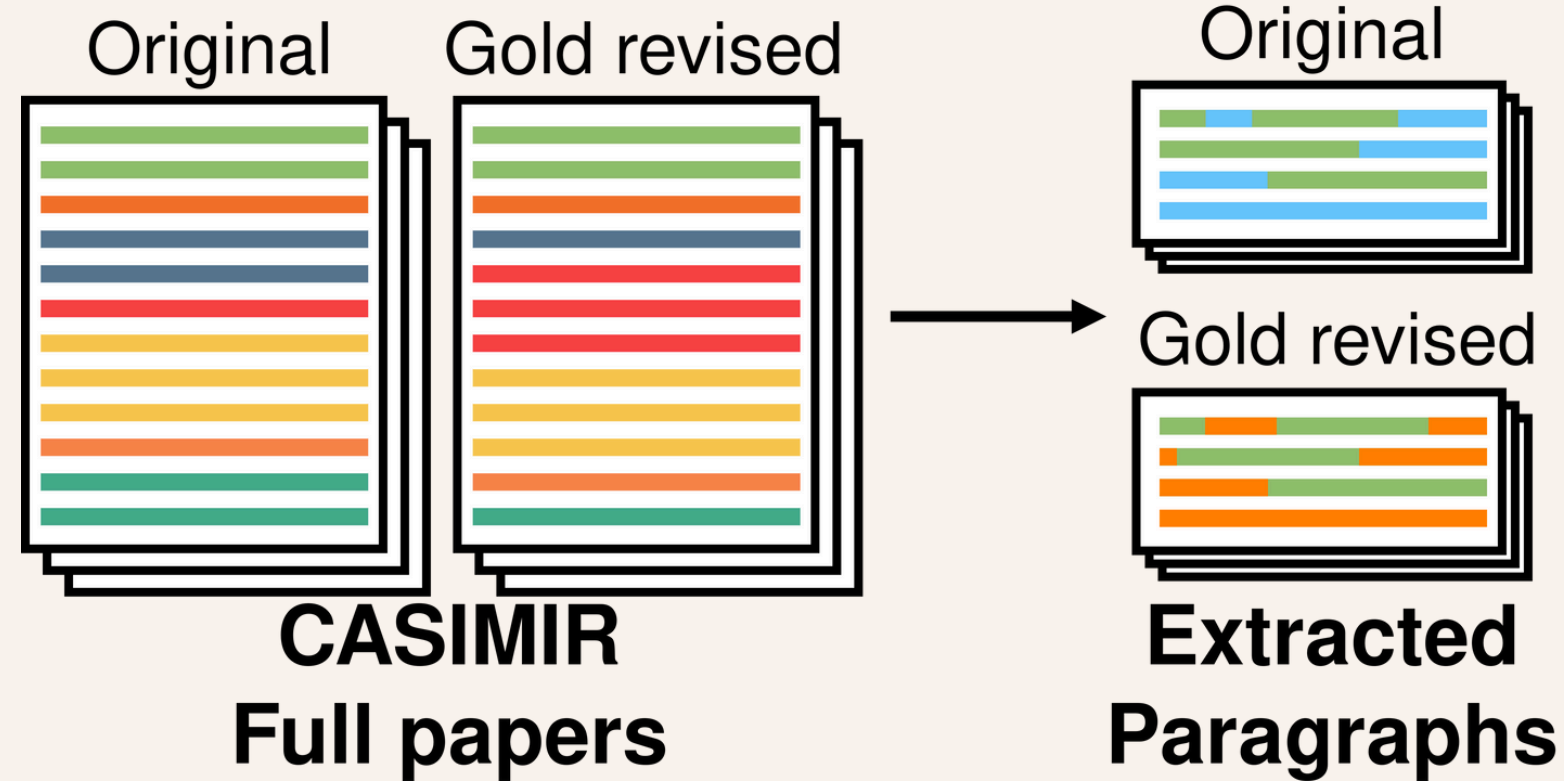
ParaRev Data pipeline – Annotation

Manual annotation



Label

Instruction



Original paragraph

[...] **Nevertheless, challenges exist for** developing deep learning-based models to predict mutational effects on protein-protein **binding. The major challenge is the scarcity of experimental data — only** a few **thousands of** protein **mutations** annotated with **the change** in binding **affinity** are publicly available (Geng et al., **2019b**). **This hinders** supervised learning **as the insufficiency of training data tends to cause over-fitting.** [...]

Gold revised paragraph

[...] **However,** developing deep learning-based models to predict mutational effects on protein-protein **binding is challenging due to the scarcity of experimental data. Only** a few **thousand** protein **mutations,** annotated with **changes** in binding **affinity,** are publicly available (Geng et al., **2019b**), **making** supervised learning **challenging due to the potential for overfitting with insufficient training data.** [...]

Manual Annotation

Revision Instruction

Rewrite this paragraph for better readability.

Intention label

Rewriting_medium

Example of data in ParaRev

Annotation

10 annotators

-  2 professors,  3 PhD students,  and 5 master's students
- not native from English
- specialized in the NLP domain
- experienced in reading and writing academic papers

Agreement

73.32% are double annotated
≈ 1.2 labels/paragraph

Krippendorff's alpha

0.50

Paragraphs sharing at least one label

75.32%



ParaRev Description

48,203 paragraphs in total from 16,664 pairs of revised articles

641 annotated paragraphs (470 with cross annotation)

Heavily revised papers

>19 paragraphs revised



218 paragraphs

Moderately revised papers

4-5 revised paragraphs



213 paragraphs

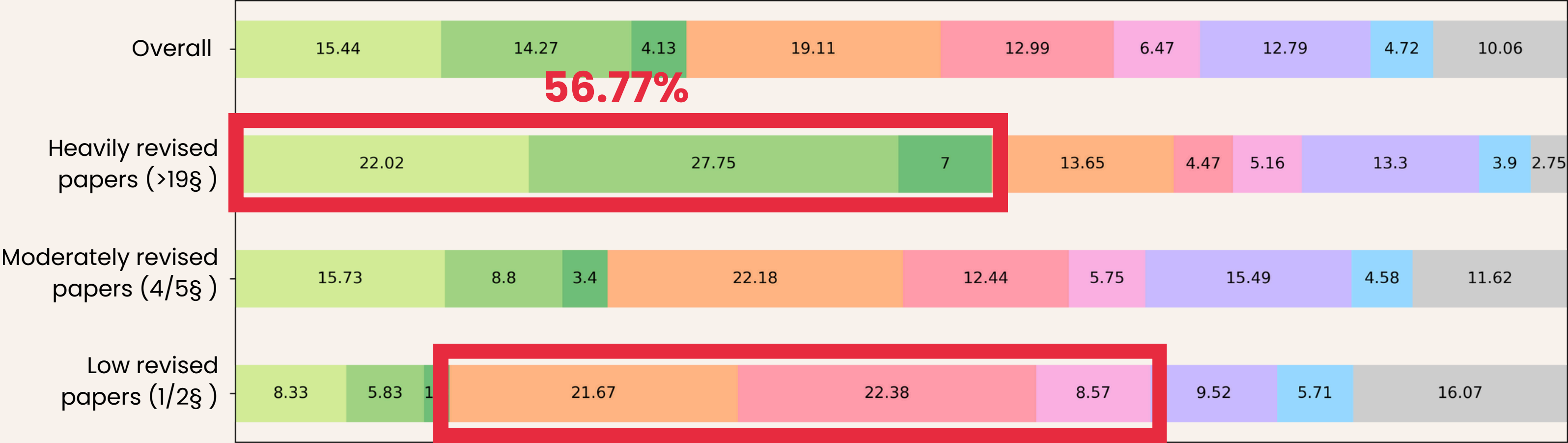
Low revised papers

1-2 revised paragraphs



210 paragraphs

ParaRev test set – Distribution



52.62%

Instructions' distribution

# instructions	0	1	2
# paragraphs	327	56	258

Evaluation set

- Rewriting light
- Rewriting medium
- Rewriting heavy
- Development
- Content addition
- Content substitution
- Concision
- Content deletion
- Unusable

Chapter recap

Contributions

- **CASIMIR**, a large dataset for scientific text revision
- **ParaRev**, a dataset of revised paragraphs, with an evaluation split annotated with revision instructions

Perspectives

- **Automatically annotate** the train data
- **Fine tune** a specialised small open source model

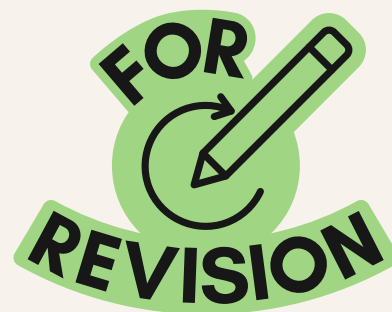
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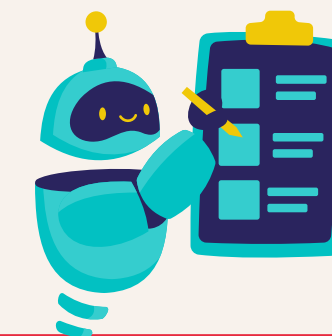
Statistics

III– Identifying Reliable Evaluation Metrics

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Metrics recommendation



Identifying Reliable Evaluation Metrics for Scientific Text Revision

1st author, ACL 2025 (Main) + CORIA TALN 2025

[*]–PLUIE : Personalisable metric in Llm Used for Improved Evaluation

2nd author, Collaboration with IRISA, Submitted to ARR 2025



CORIA-TALN 2025
MARSEILLE



Metrics used on STR

Similarity based on:

- n-grams

- embeddings

- LLM

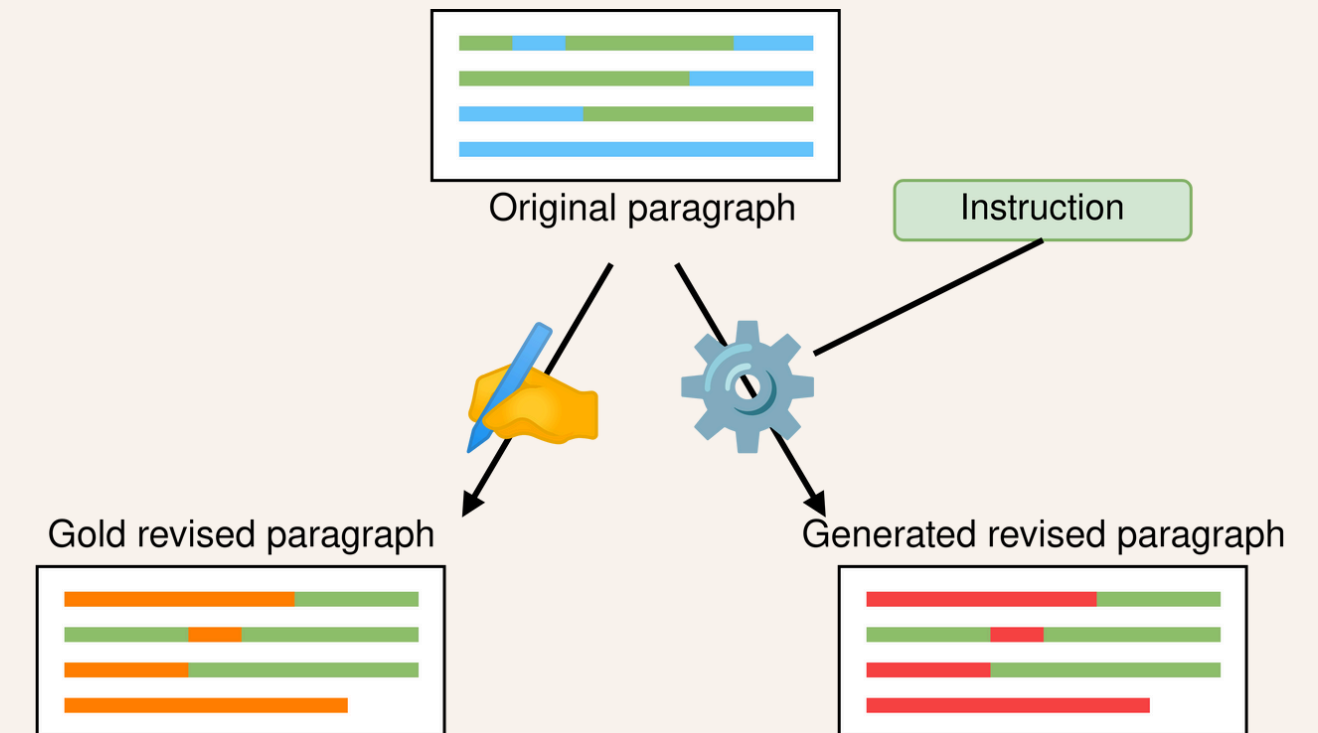
BLEU
ROUGE
METEOR
SARI
GLEU
BERTScore
LLM-as-a-judge

Input:

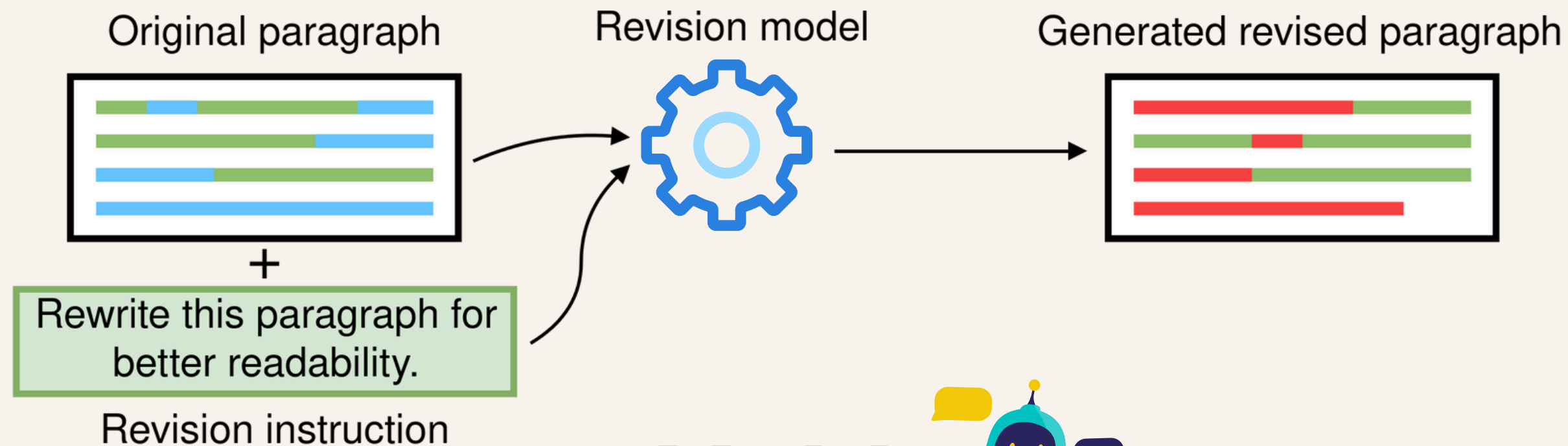
- Gold + Generated

- Original + Gold + Generated

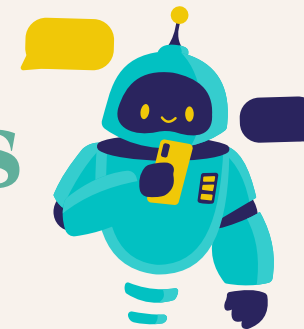
- Original + Gold + Generated + Instruction



Revision generation



Models



CoEdit-xl (3B)
(fine tuned T5)



Mistral AI

Mistral-7B-Instruct-v0.2



Meta

Llama-3-8B-instruct
Llama-3-70B-instruct



OpenAI

GPT4o-mini
GPT4o



*no edits
control approach*

Performances of Revision Models

Revision Model ↓ / Metric →	BLEU	ROUGE-L	METEOR	GLEU	SARI	Bertscore
no edits control approach	66.00	78.30	83.80	25.78	60.63	95.95
CoEdIT-XL	50.24	67.46	66.66	23.84	39.60	93.90
Mistral-7B-Instruct-v0.2	27.77	50.79	54.02	15.38	31.63	92.14
Meta-Llama-3-8B-Instruct	41.66	62.07	62.00	25.78	39.33	93.53
Meta-Llama-3-70B-Instruct	46.78	65.61	67.20	30.31	42.74	93.90
GPT4o-mini	<u>51.68</u>	<u>69.54</u>	<u>72.70</u>	32.67	<u>45.06</u>	<u>94.80</u>
GPT4o	49.34	68.20	69.88	<u>31.35</u>	43.54	94.45

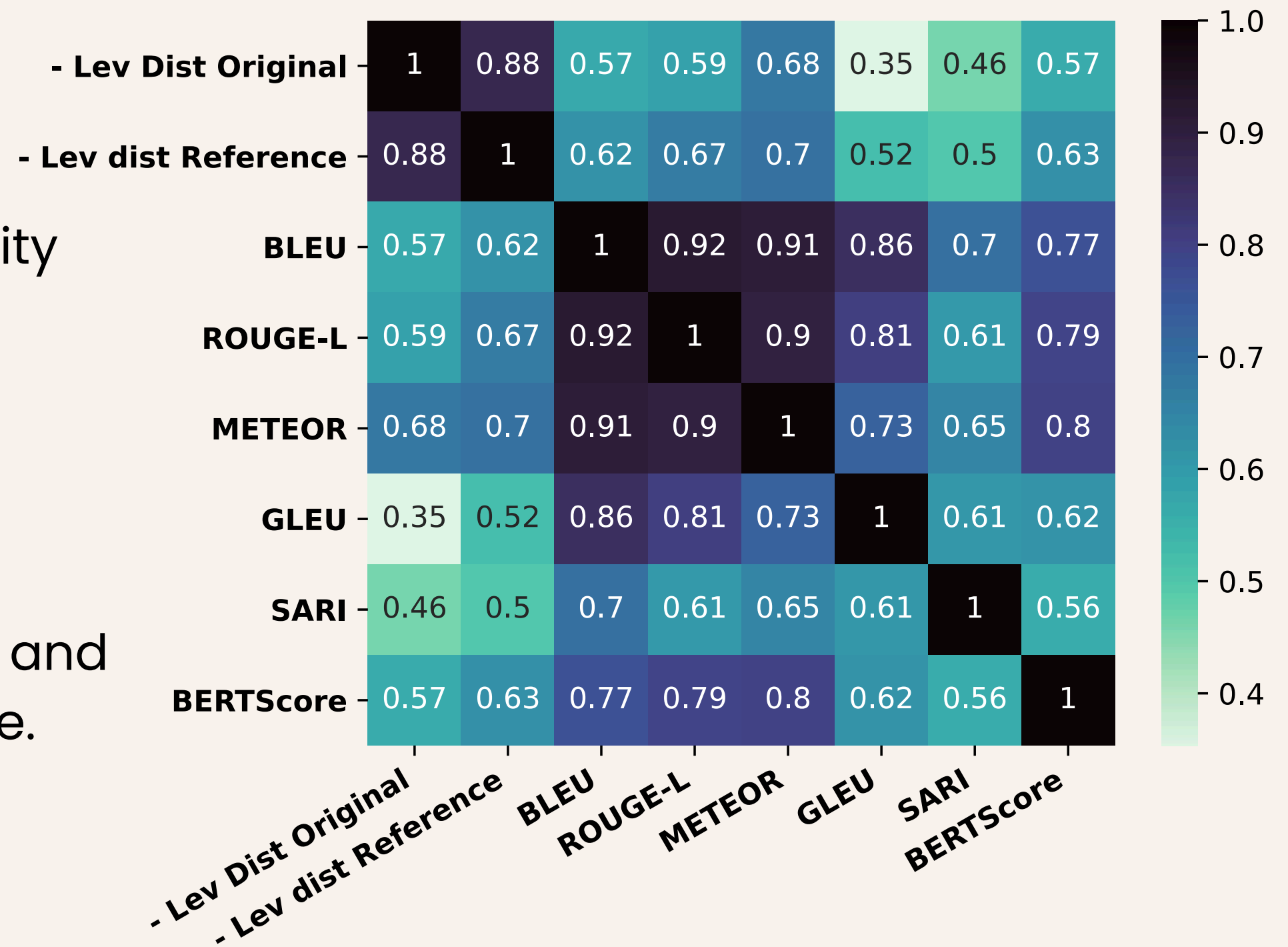
Limitations of Similarity-based Metrics

1. Most metrics are highly correlated,
providing redundant information

2. Metrics only capture surface similarity

3. Substantial revisions are penalised

SARI and GLEU stand out as exceptions, and
are the only ones to consider the source.



Exploring Alternative Evaluation Approaches

 **Goal:** Define a set of complementary and reliable metrics to evaluate paragraph revision



1. Manual pairwise evaluation of generated revisions by different models

3 levels:

1. **Relatedness**
2. **Correctness**
3. **Preference**



2. Identify candidates metrics

Related NLP Domains

Input: Og + Gen

Dale-Chall

BETS

BLANC

ParaPLUIE

LLM-as-a-judge

Input: Og + Inst + Gen

LLM-choice

LLM-likert

InstructLUIE

3. Align human judgment and automatic evaluation

Exploring Alternative Evaluation Approaches

2. Identify candidates metrics

Related NLP Domains

(Reference-free)

Input: Original + Generated

Dale-Chall: Readability (*Word counts*)

BETS: Text simplification
(*embedding-based*)

BLANC: Summarization (*bert-based*)

ParaPLUIE: Paraphrase detection
(*LLM-based*)

LLM-as-a-judge

Input: Original + Instruction + Generated

Using LLM's output

LLM-choice

(Yes/No + Pairwise comparison)

A or **B**

LLM-likert

(Generating Scores)

1 - 2 - 3 - 4 - 5

Using token's probabilities

InstructLUIE

(ParaPLUIE with instruction
following prompt)

$P(\text{"No"}) - P(\text{"Yes"})$

Results

Judge	Pair acc.	V	K
LLM choice (<i>Llama 3 70B</i>)	<u>0.592</u>	0.279	<u>0.297</u>
LLM likert (<i>GPT-4o</i>)	0.541	<u>0.281</u>	0.276
InstructLUIE (<i>Llama 3 70B</i>)	0.607	0.311	0.317
ParaPLUIE (<i>Mistral 7B</i>)	0.551	0.241	0.218
BETS	0.492	0.152	0.127
BLANC	0.357	0.117	-0.080
Dale-chall	0.512	0.226	0.159
Bertscore	0.445	0.161	0.034
SARI	0.465	0.184	0.071
GLEU	0.504	0.193	0.138
ROUGE-L	0.414	0.179	-0.013
<i>Random weighted</i>	<i>0.358</i>	<i>0.030</i>	<i>-0.003</i>

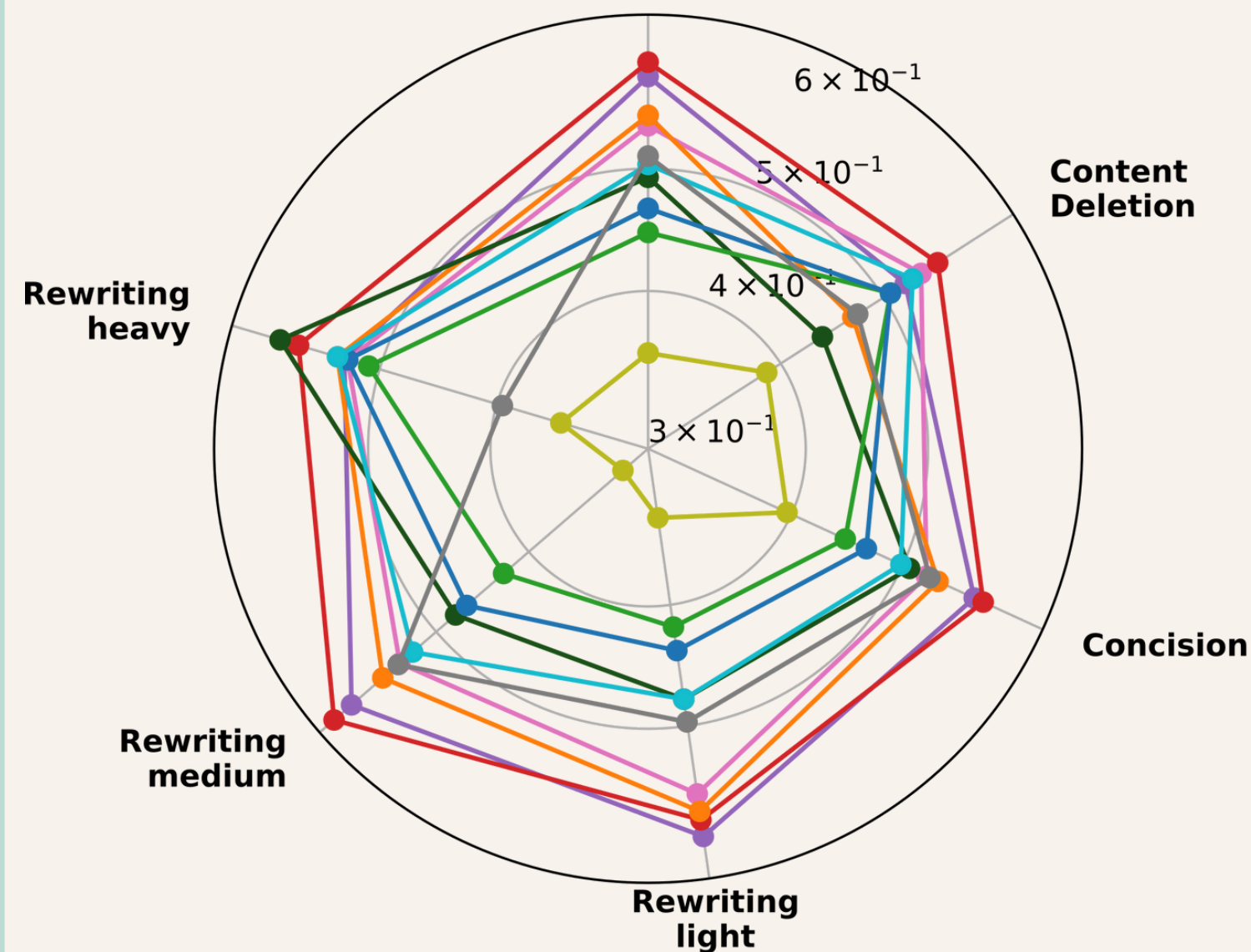
Alignment of automatic metrics with human judgements

Performance by Aspect

By Revision Category

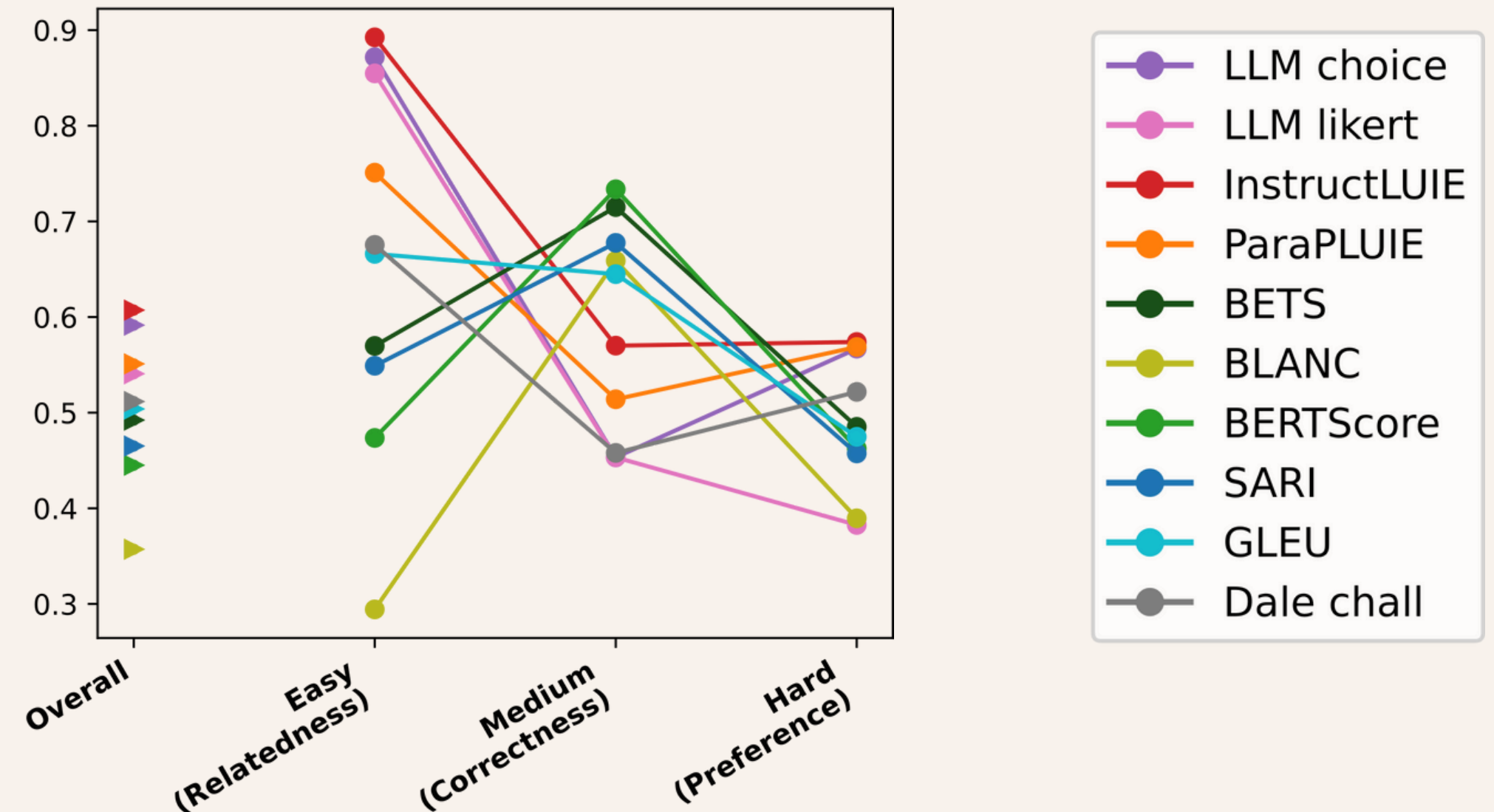
Pairwise accuracy

Overall



By Difficulty

Pairwise accuracy



Proposed Alternative Evaluation

1. **Instruction following**: LLM-as-a-judge
2. **Similarity to gold**: SARI or GLEU
3. **Meaning-preservation**: ParaPLUIE

Chapter recap

- Traditional similarity metrics alone **fail to accurately evaluate** text revision
- Adapted an existing metric on our task: **InstrucLUIE**
- An **alternative evaluation method** of 3 complementary metrics:
 - **Instruction following**: LLM-as-Judge (*InstrucLUIE* or *LLM-choice*)
 - **Similarity to gold**: SARI or GLEU
 - **Meaning preservation**: ParaPLUIE with original paragraph
- **ParaReval, a dataset** of human pairwise evaluations of generated revisions

CONCLUSION

Contributions

Definition of the text revision task at **paragraph-level**,
with personalised **revision instructions**

Experimental framework comprising

New resources dedicated to STR composed of three complementary datasets



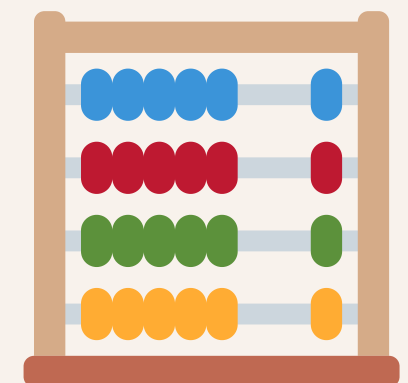
CASIMIR, a large scale corpus of **15k scientific articles** with revisions, aligned at **sentence-level**

Pararev, a dataset of **48k revised paragraphs** with an evaluation subset of **641 manually annotated** paragraphs

ParaReval, a dataset of **human pairwise evaluations** of generated revisions

Highlighted a set of **recommended reliable metrics** for evaluating revision systems through an extended analysis

Introduced a new metric, **InstructLUIE**



Future directions

LLM4Science

- **Models as research assistant:** write from scratch, propose ideas, review

Ethics

- **Ethical considerations:** intellectual property, impact on language learning, authorship, sovereignty on data, energy cost
- Use **small open models**